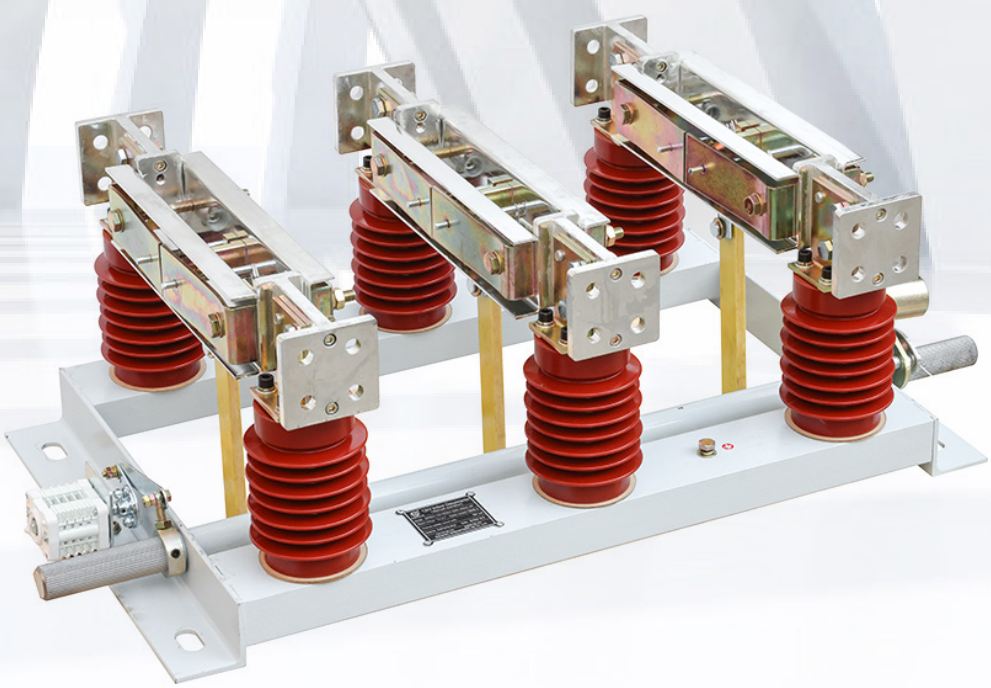
Installation and Operation Manual

GreenPower

# GN22-12(C)(D) Series

# Indoor High Voltage Disconnector





# About Us»

GreenPower is a manufacturer of medium- and low-voltage switchgear and components, with in-house engineering, manufacturing and testing capabilities. Our manufacturing entity, Zhejiang GreenPower Electric Co., Ltd., was established in 1992. International business and export operations are handled by Zhejiang GreenPower I&E Co., Ltd.

Our product portfolio covers medium-voltage switchgear from 12 kV to 40.5 kV, including air-insulated metal-clad switchgear (AIS), cubicle-type gas-insulated switchgear (C-GIS), ring main units (RMU), vacuum circuit breakers, and a broad range of switchgear components and accessories. We also manufacture low-voltage withdrawable switchgear systems with ratings up to 6300 A.

Engineering and manufacturing are central to what we do. Our production and testing capabilities include automated sheet-metal processing, CNC laser cutting, 3D five-axis laser welding for stainless-steel gas tanks, automatic helium leak detection, and a fully shielded partial-discharge test room. These capabilities support consistent manufacturing quality from enclosure fabrication through assembly and final testing.

For customers moving toward lower-impact power distribution systems, GreenPower offers SF6-free solutions including dry-air insulated RMUs and nitrogen-insulated C-GIS, alongside conventional gas-insulated solutions where required. These technologies help customers respond to increasingly stringent environmental requirements for medium-voltage switchgear.

We also support project-specific engineering based on customer drawings and specifications, including non-standard dimensions, special voltage and frequency requirements, high-altitude applications, customized component selection and other application-specific configurations.

Our goal is simple: to be the switchgear manufacturer our customers return to — because the equipment is properly engineered, reliably manufactured and built for long-term service.

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***Warranty*** **18 months after shipment date or 12 months after take-over, whichever comes first.**

## 01 Product overview

GN22-12(C)(D) series indoor high-voltage isolating switch is used in power systems with rated voltage of 12kV, 50Hz and below, to connect, disconnect or convert lines when there is voltage but no load.

## 02 Environmental conditions for use

- Ambient air temperature: -15℃ ~ 40℃ , daily average temperature ≤ 35℃ .
- Altitude: ≤ 1000m.
- Temperature conditions: Daily average relative humidity ≤ 95%, daily average water vapor pressure ≤ 2.2kPa;
  Monthly average relative humidity ≤ 90%, monthly average water vapor pressure ≤ 1.8kPa.
- Earthquake intensity: ≤ 8 degrees.
- No fire, explosion hazard, severe pollution, chemical corrosion and severe vibration.

**Note: If the product usage conditions exceed the above provisions, the user and the company shall negotiate and determine.**

## 03 Main technical data

| Model | Rated voltage (kV) | Rated current (A) | 4s Thermal stability current (effective value) (kA) | Dynamic stability current (peak) kA |
|---|---|---|---|---|
| GN22-12(C)(D)/2000A | 12 | 2000 | 50 | 100 |
| GN22-12(C)(D)/2500A | | 2500 | 50 | 125 |
| GN22-12(C)(D)/3150A | | 3150 | 50 | 125 |
| GN22-12(C)(D)/4000A | | 4000 | 50 | 125 |
| GN22-12(C)(D)/5000A | | 5000 | 100 | 250 |
| GN22-12(C)(D)/6300A | | 6300 | 100 | 250 |
| GN22-12(C)(D)/8000A | | 8000 | 100 | 250 |

## 04 Structural features

This product is a high-current isolating switch with a three-phase common bottom knife structure. It adopts a mechanical locking prototype with a two-step action of closing and contacting, achieving wear-free contact, small operating force, high contact pressure, and good dynamic thermal stability.

## 05 Working principle

GN22-12(C)(D) type is divided into flat-mounted type, through-wall type, flat-mounted grounding type, through-wall grounding type. The grounding type is further divided into moving contact side grounding, static contact side grounding and moving and static contact double grounding. This switch is a three-phase common base frame knife switch. As shown in the figure, it can be divided into a base frame, an insulator and a conductive part. The base frame is composed of a rotating shaft and a frame welded with a bent plate and an angle steel. Six insulators are fixed on the base frame, and each two form a pole. A three-phase operating insulator is installed on the rotating shaft to operate the three-phase knife opening and closing movement. The conductive part consists of a contact, a knife and a contact seat, which is compressed by a spring.

When opening, the operating crank arm drives the rotating shaft to rotate, so that the operating insulator pushes the knife upward to separate the knife and the contact. The knife rotates around the head, and the knife also moves upward to the opening position driven by the operating insulator.

When closing the circuit breaker, the operating arm drives the shaft to rotate, causing the operating insulator to pull the knife downward until the knife reaches the rotation limit position (but not locked yet), and then the mechanism drives the main shaft to continue rotating about 10° (a "dong" sound can be heard), at which time the knife and contact are locked.

**Note: If the switch is not locked, it cannot be put into operation, otherwise it will cause the switch to overheat and burn.**

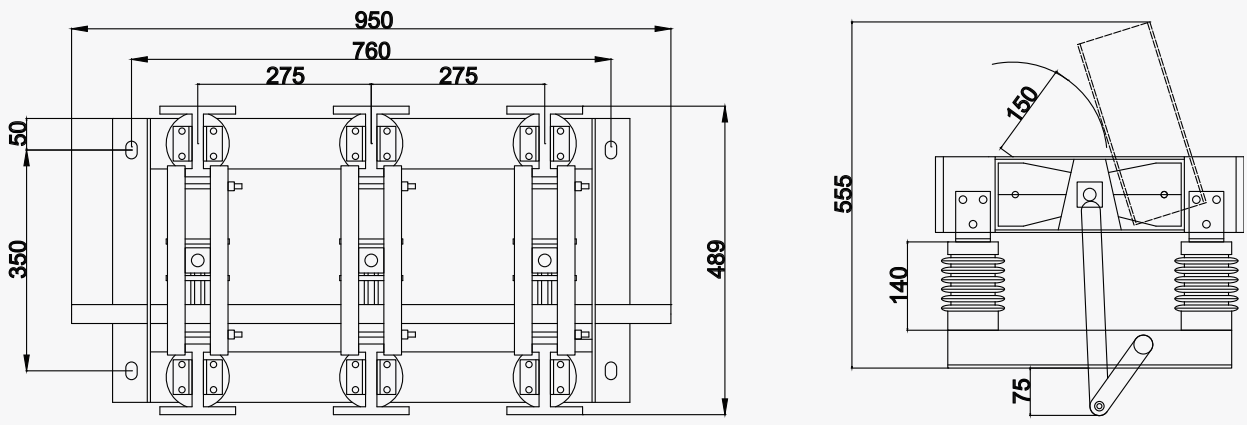
**Matching mechanism: Equipped with CS6-1, CS6-2 and other manual operating mechanisms, and electric operating mechanisms can also be used.**

**This series of switches are operated by CS6-2 manual operating mechanism or electric operating mechanism, and can be installed on the left or right side of the disconnector.**

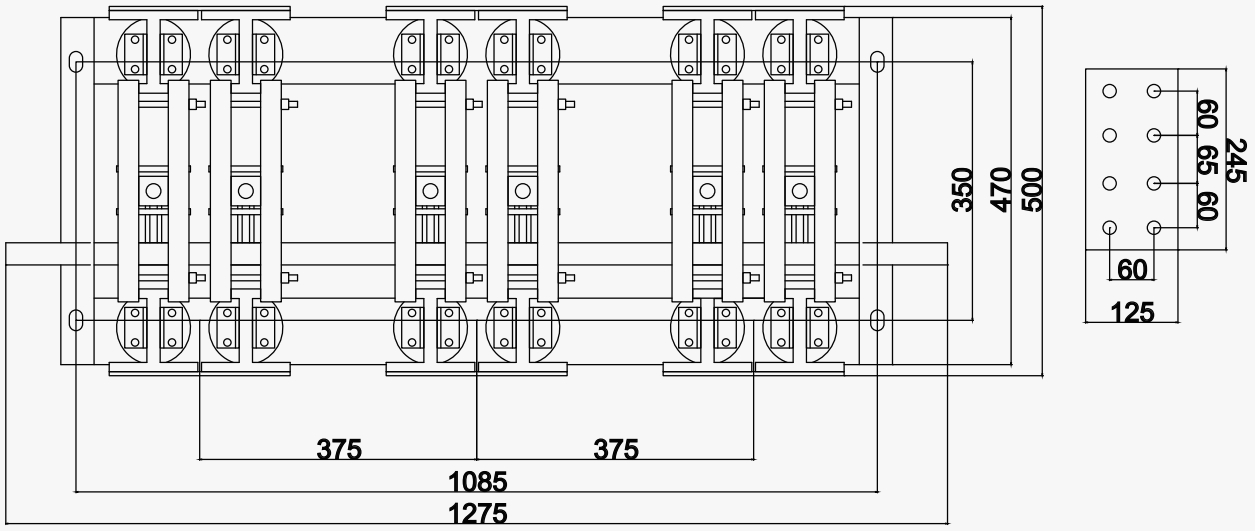
## 06 Packaging, transportation and storage

The packaging method is a wooden box. The switch body, random accessories, and documents (product certificate, instruction manual) are all packed in one wooden box. After the product arrives at the destination, if it is not installed immediately, it must be placed in a rainproof, dry, and ventilated place. If it is stored for a long time, it should be checked regularly.

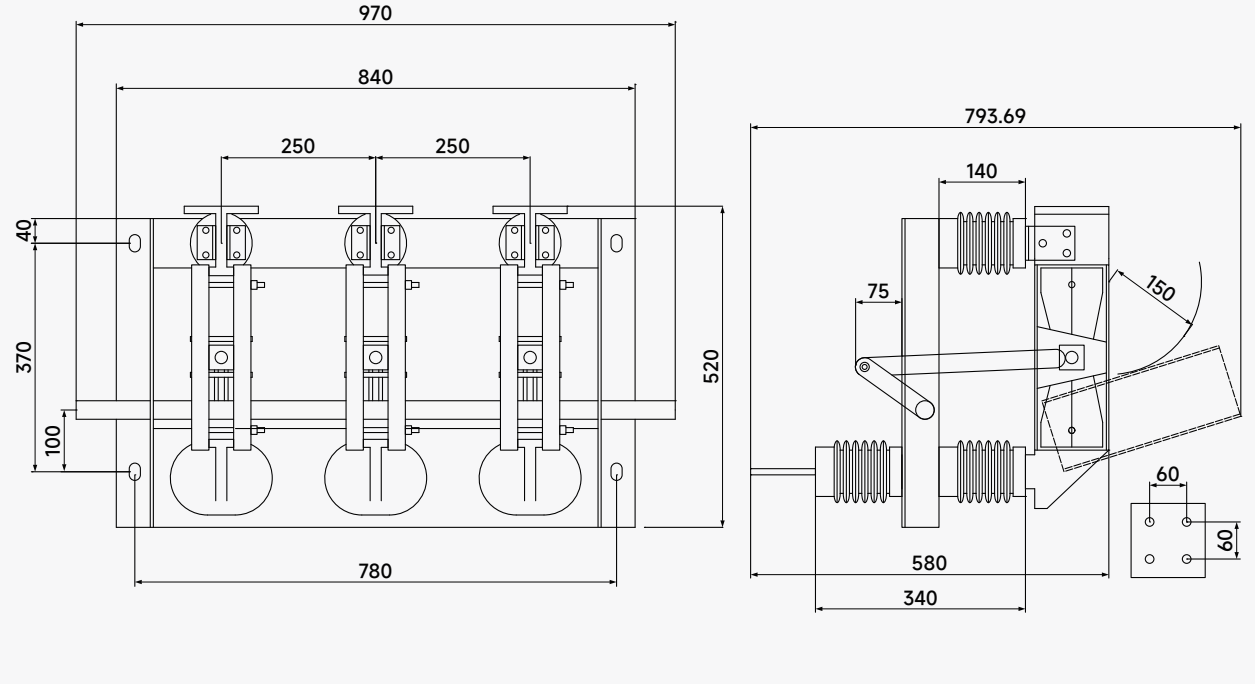
## 07 Appearance and installation dimension drawing



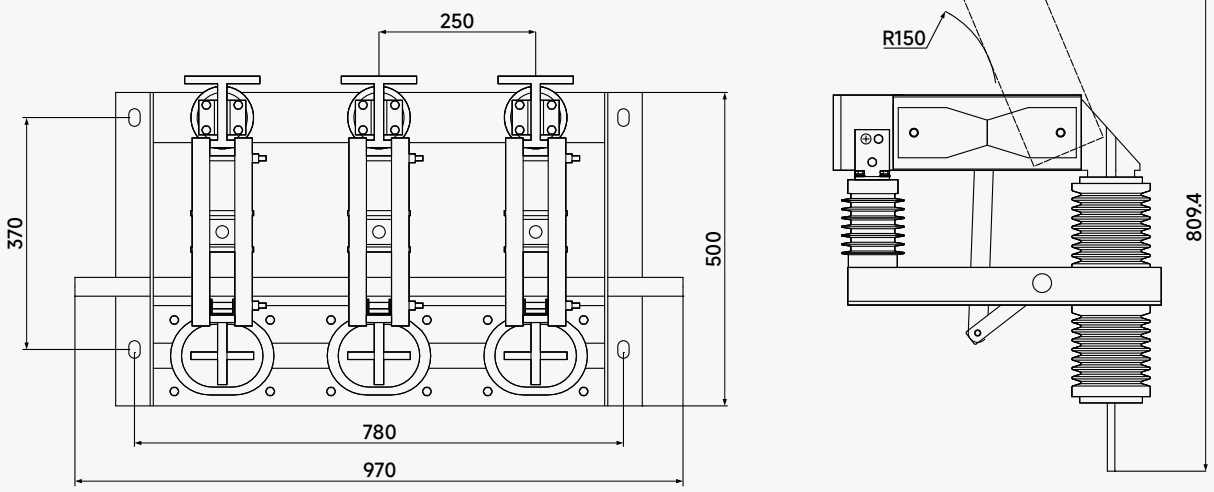

**Drawing 1** GN22-12/3150A-275



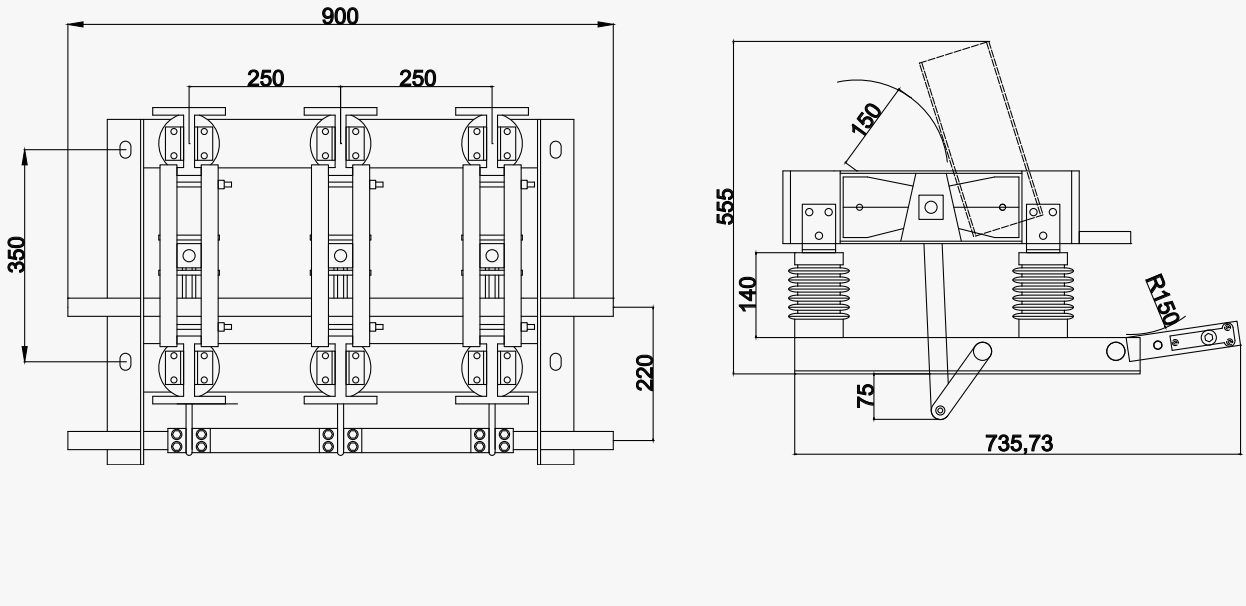

**Drawing 2** GN22-12/(6300&8000)-375



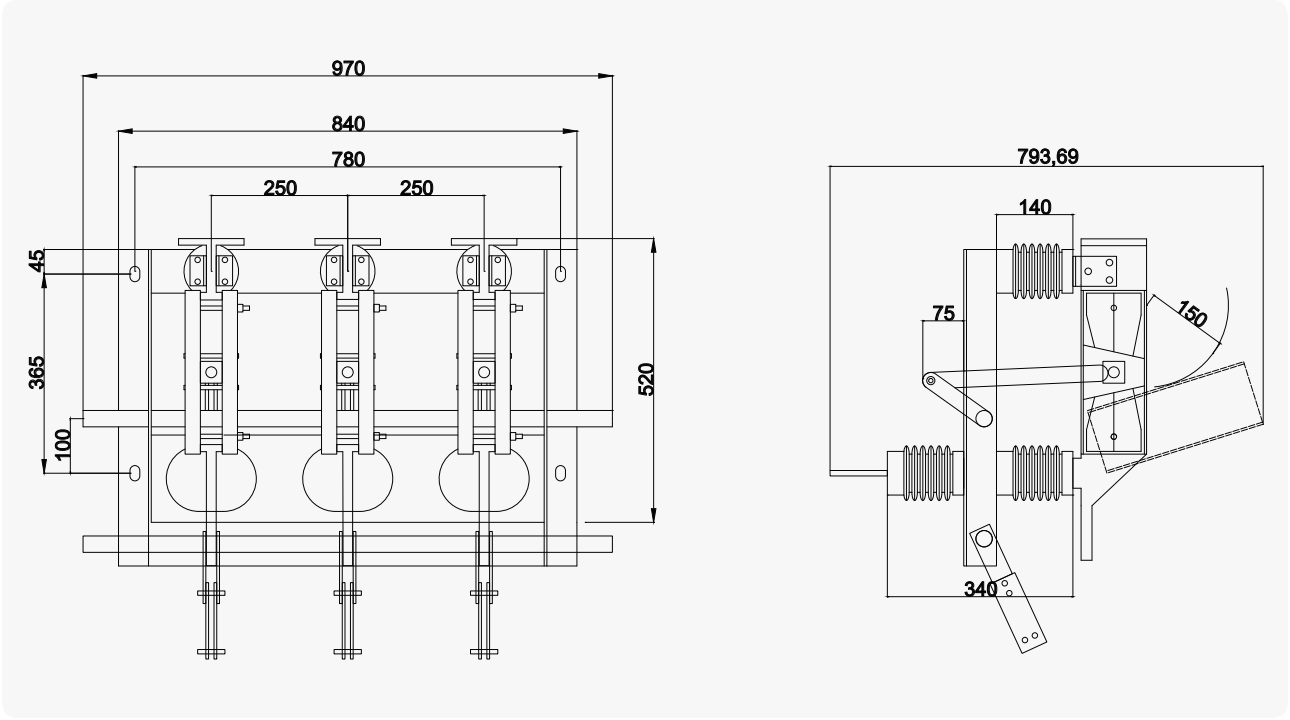

**Drawing 3** GN22-12C/2500-250



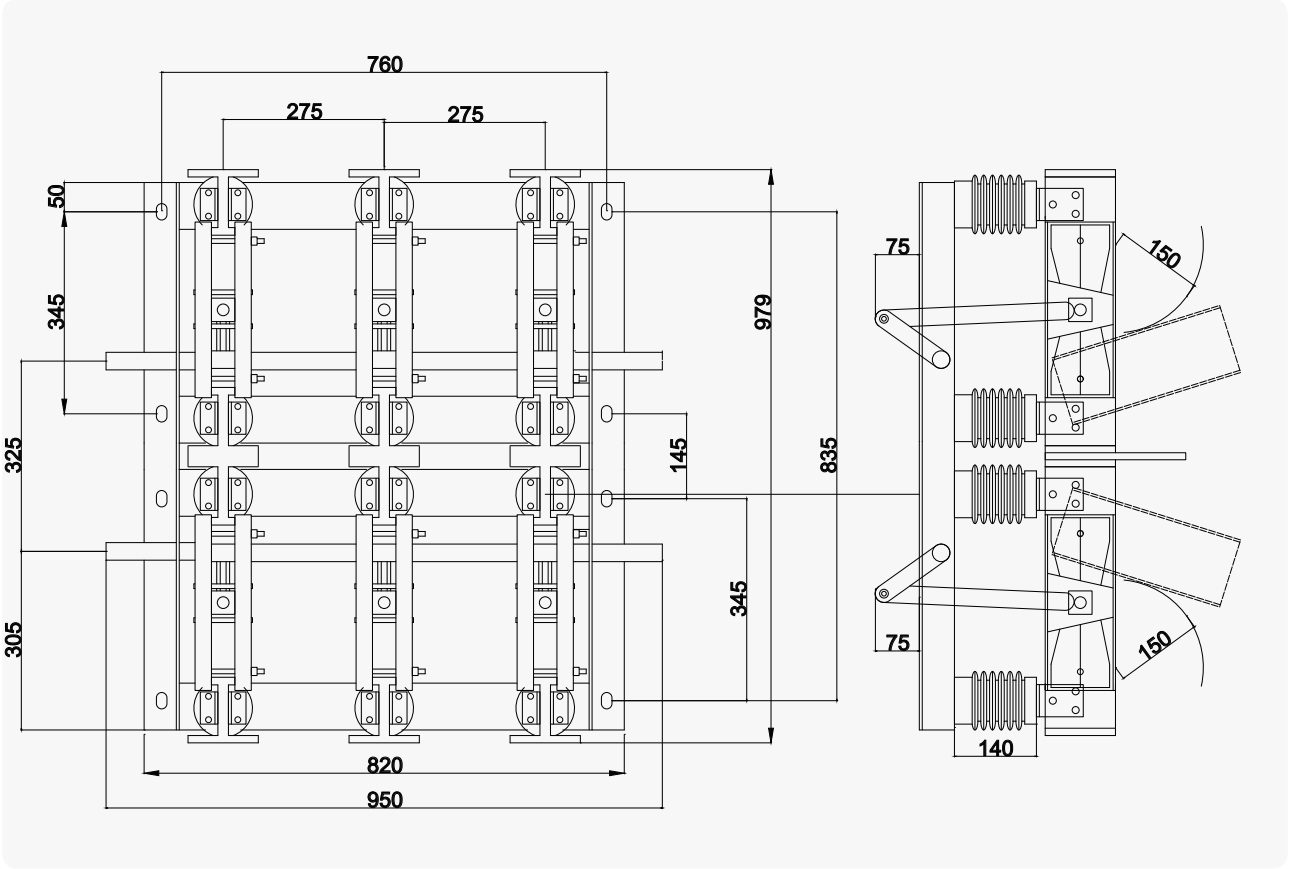

**Drawing 4** GN22-12C/3150-250



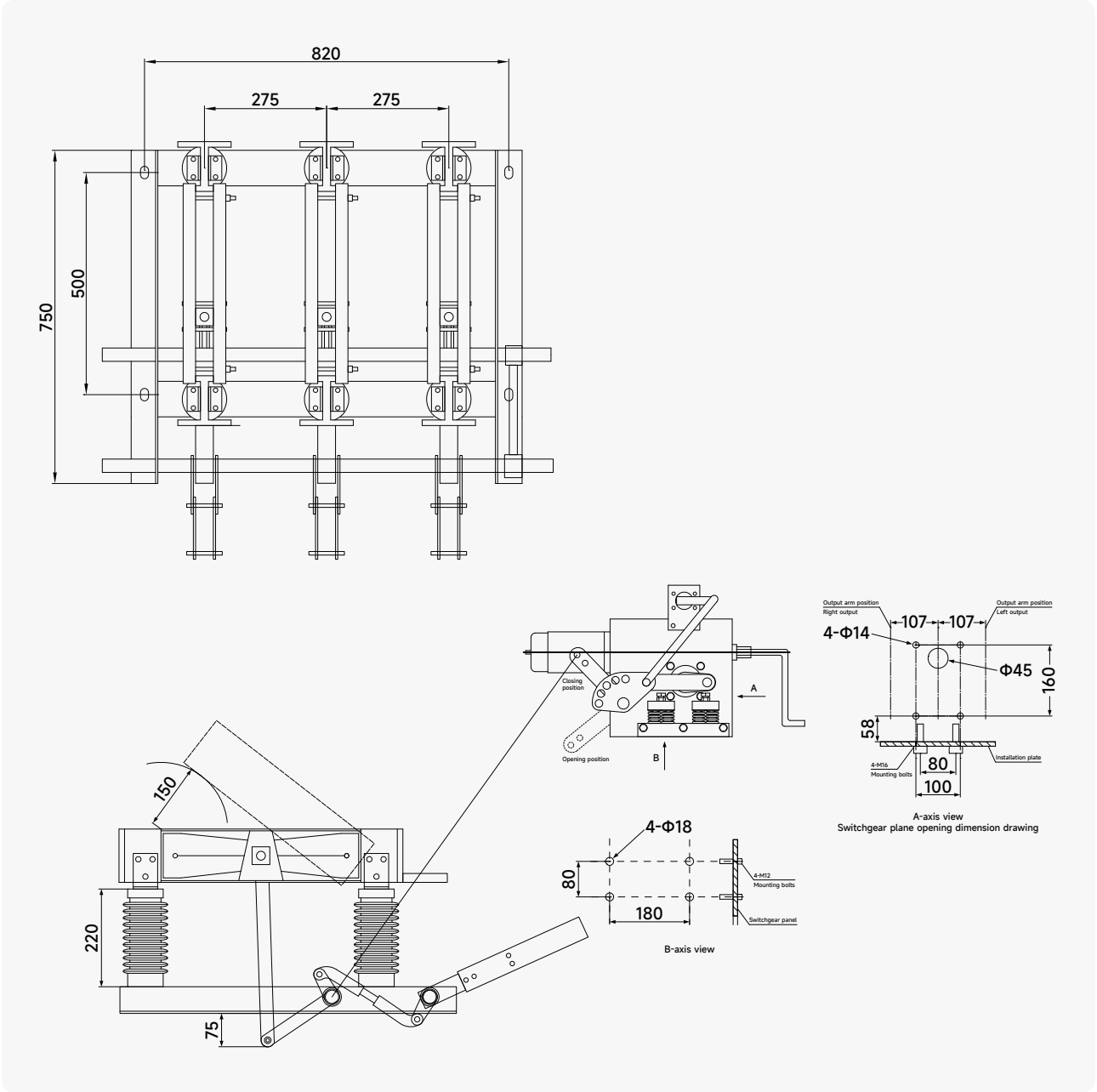

**Drawing 5** GN22-12D/3150-250




**Drawing 6** GN22-12CD/2500-250




**Drawing 7** GN22-12ST/3150-250




**Drawing 8** GN22-12ST/3150-250 with electric mechanism (split installation)

## 08 Ordering instructions

- When ordering this switch, please specify the model specification, rated voltage, rated current and other technical data.
- If you have special requirements, you can negotiate with the manufacturer to resolve them.

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**Zhejiang Greenpower Electric Co., Ltd**
**Zhejiang GreenPower I&E Co., Ltd**
**Professional Power Equipment Provider**

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We reserve the right to make modifications following technical developments to the products.

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